

# Getabalew Manegerew Sewbekele

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## EDUCATION

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Addis Ababa University  
Bachelor of Science in Physics

Addis Ababa, Ethiopia  
#1 Ranked in Class of 2026

- **Cumulative GPA:** 4.00/4.00
- **Core Theory:** Classical Mechanics, Quantum Mechanics I & II, Electrodynamics I & II, Statistical Mechanics, Mathematical Methods I & II
- **Relevant Courses:** Astronomy and Astrophysics I & II, Computational Physics, Applications of Computational Physics, Nuclear Physics

## RESEARCH EXPERIENCE

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Addis Ababa University

Addis Ababa, Ethiopia

Senior Thesis: Direct Collisional Method of Molecular Dynamics Simulation

Feb – Jun 2026

- Proposed and implemented (Fortran 90) a novel hybrid DSMC–MD algorithm for rarefied hard-sphere gas systems, preserving full particle histories at DSMC-level computational efficiency.
- Validated the model by demonstrating ergodic convergence to the Maxwell–Boltzmann distribution and benchmarking energy, heat capacity, and mean free path against theoretical predictions.
- Extended the model to ionizing radiation propagation in gases; analyzed outputs using NumPy and Matplotlib.

Addis Ababa University

Addis Ababa, Ethiopia

Undergraduate Researcher & Team Lead — Atmospheric Muon Angular Flux

Mar – Aug 2025

- Led a 3-person team investigating relativistic corrections to atmospheric muon trajectories in Earth's magnetic field; diagnosed and resolved a GPS firmware fault, restoring full detector data acquisition.
- Built a Python Monte Carlo simulation of muon trajectories, reproducing experimental angular distributions; optimized detector operating voltage via the coincidence-plateau method.
- Authored a faculty-approved proposal demonstrating the necessity of relativistic over Newtonian treatment.

## INDEPENDENT ACADEMIC COURSEWORK

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- **General Relativity:** Organized and lead a remote, self-paced mini-course with students from Addis Ababa University and other Ethiopian universities, following Schutz's *A First Course in General Relativity*; completed Special Relativity, tensors, manifolds, and currently progressing spacetime geometry.
- **MIT OpenCourseWare:** Self-study of Linear Algebra, Multivariable Calculus, Classical Mechanics, Electromagnetism, and Quantum Mechanics.
- **Computational Modeling:** Completed UC Berkeley's *Python4Physics* curriculum; applied methods from *A Student's Guide to Python for Physical Modeling*.
- **Machine Learning & Quantum Computing:** Currently self-studying Scikit-learn and Qiskit.

## TEACHING & PROFESSIONAL EXPERIENCE

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Addis Ababa University

Python Summer Camp Instructor

Jul 2026 – Present

- Lead lab and problem-solving sessions; deliver lectures on algorithm development and Python syntax for high school students.

**Addis Ababa University**    *Teaching Assistant — Statistical Mechanics*    *Oct 2025 – Jan 2026*

- Delivered weekly recitation sessions on statistical mechanics; developed Monte Carlo-based computational visualizations of Boltzmann distribution convergence.

**Ethiopian Conformity Assessment Enterprise**    *Radiation Spectrometry Intern*    *Jun – Sept 2025*

- Calibrated HPGe and alpha spectrometry systems for national radiation compliance; designed a relational MS Access database to modernize the national dosimetry program.

## SKILLS & TOOLS

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- **Programming:** Python (NumPy, Pandas, Matplotlib), Fortran 90 (RK4 solvers, numerical integration/differentiation, physical modeling), C++ (numerical algorithms)
- **Software & Hardware:** Mathematica, LaTeX, Gnuplot, MS Access, Linux/Unix; cosmic ray detectors, HPGe spectrometers

## HONORS & SCHOLARSHIPS

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- **EPSNA Merit-Based Scholarship**, Ethiopian Physics Society in North America (2024, 2025); Virtual Graduate-Level Summer School participant (Aug 2025)
- **Ethio-China Friendship Scholarship** (2024–2025) and **Mitsubishi Scholarship Award** (2023–2024), both merit-based academic excellence awards

## ADDITIONAL ACTIVITIES

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- **Co-Founder, FOCAL Physics Club** — organized bimonthly research seminars and faculty panels bridging undergraduate study with research pathways.
- **Invited Speaker** — guest lecture on thermodynamics (AAU, Dec 2025) and seminar on special relativity & electromagnetism (FOCAL Club, Apr 2023).
- **Keynote Speaker**, “Physics Opportunities & Pathways,” Sophomore Welcoming Event, Addis Ababa University (Oct 2025)
- **Class Representative** (2022–2026) — built a digital pipeline for course material distribution and faculty–student communication.